Nutrition & Health Tracker - Pre-PRD Feature Baseline

Feature and decision baseline captured before formal PRD generation.

1. Product Direction

The application is a personal-first nutrition and health tracker, but it should be designed as a proper multi-user application from the beginning.

Initial expected usage is small, but the architecture should not assume that only one user will ever use it.

Primary goals:

- Accurate nutrition tracking.
- Strong support for Indian foods and Indian dietary patterns.
- Medication and supplement tracking.
- User-controlled nutrition targets.
- Historical nutrition analysis.
- Extensible architecture so advanced functionality can be added later.
- Tracking/reference application, not a medical recommendation system.

Accuracy is more important than minimizing the number of steps required to log food.

2. Phase 1 - Core Tracking Application

Authentication and Users

- User registration.
- Login.
- Logout.
- Password reset.
- Secure password storage.
- Session/token management.
- Multiple users.
- Strict separation of each user's data.

Initial authentication can be email/password.

Google login and additional authentication providers are not required in Phase 1.

3. User Profile

User profile should store information required for nutrition calculations.

Example fields:

- Name.
- Date of birth / age.
- Sex.
- Height.

- Weight.
- Activity level.
- Goal/preferences.
- Timezone.

The profile should also contain nutrition and application preferences.

4. Daily Reset Time / Tracking Day

The application should not assume that a user's nutrition day always starts at midnight.

The user should be able to configure a Daily Reset Time.

Default: 12:00 AM.

Example: If a user configures 4:00 AM, food eaten at 1:00 AM should count toward the previous tracking day.

Therefore the application should introduce the concept of a Tracking Day.

A Tracking Day is calculated using:

- Timestamp.
- User timezone.
- User-configured reset time.

All daily food/nutrition aggregation should use Tracking Day rather than simply the calendar date.

Medication/supplement schedules should primarily use actual timestamps so changing the nutrition reset time does not unexpectedly modify medication schedules.

5. Nutrition Requirements and Targets

System should calculate nutrition reference requirements using profile information where relevant.

Possible inputs include:

- Age / life stage.
- Sex.
- Height.
- Weight.
- Activity.
- Goal.

Not every nutrient should use a body-weight formula.

Where appropriate, established nutrition reference values should be used.

Indian authoritative sources should be preferred where possible, such as:

- ICMR.
- NIN.
- Indian food composition/reference datasets where licensing permits their use.

System should distinguish between Reference Nutrition Requirement and Active User Target.

Example: Protein reference = 130 g; Protein active target = 150 g.

These must not be treated as the same value.

6. Nutrition Target Review and Overrides

Nutrition targets should not silently change.

When targets are initially calculated, the user should be able to:

- Review all proposed values.
- Accept all.
- Accept individual nutrients.
- Manually edit values.
- Reject proposed values.

If the user's profile later changes, the application can recalculate requirements.

However, recalculation should create proposed new values.

The active targets should only change after user approval.

Example: Current protein target = 130 g. Proposed recalculation = 145 g. The application should show both and let the user choose whether to update.

Manual overrides should always be supported.

7. Nutrition Target History

For every target, retain information such as:

- Nutrient.
- Reference value.
- Active value.
- Unit.
- Reference source.
- Reference version.
- Calculation version.
- Whether user manually overrode the value.
- Timestamp/effective date.

This prevents target changes from becoming unexplained historical mutations.

8. Nutrients

Phase 1 should focus primarily on:

- Calories.
- Protein.
- Carbohydrates.
- Fat.

However, the underlying nutrition model should support additional nutrients from the beginning.

Examples:

- Fiber.
- Calcium.
- Iron.
- Magnesium.

- Potassium.
- Sodium.
- Zinc.
- Vitamin A.
- Vitamin C.
- Vitamin D.
- Vitamin B12.
- Folate.

The architecture should not require schema redesign later when more nutrients are displayed.

9. Configurable Tracked Nutrients

The user should be able to configure which nutrients appear prominently on the dashboard.

Possible controls:

- Select nutrient.
- Deselect nutrient.
- Select all.
- Clear all.
- Restore defaults.
- Reorder selected nutrients.

Suggested default dashboard nutrients:

- Calories.
- Protein.
- Carbohydrates.
- Fat.
- Fiber.
- Sodium.
- Calcium.
- Iron.
- Potassium.
- Vitamin C.
- Vitamin D.
- Vitamin B12.

This configuration affects the main dashboard only.

10. Full Nutrition Analysis

There should be a separate Full Nutrition Analysis screen.

This screen should show every nutrient for which nutrition data is available, regardless of what the user selected for the dashboard.

Example dashboard: Protein 108 / 130 g.

Full Analysis may additionally show calcium, iron, magnesium, vitamins, and other available nutrients.

11. Food Data Strategy

The application should avoid maintaining a massive food master database in Phase 1.

Instead, food information should primarily come from external nutrition providers/APIs.

The backend should define a provider abstraction, allowing providers to be changed or combined later.

Possible sources include:

- Open Food Facts.
- USDA FoodData Central.
- India-focused food datasets/providers where legally and technically usable.

Provider selection should consider:

- Indian food coverage.
- Nutrition completeness.
- API reliability.
- Rate limits.
- Licensing.
- Cost.
- Performance.
- Data freshness.

12. No Invented Nutrition Values

If a provider does not return a nutrient value, the application should not assume 0 unless zero is actually known.

Missing nutrients should be represented as Unknown or Not available.

Provider failures should distinguish between:

- Food not found.
- Provider unavailable.
- Rate limited.
- Invalid response.
- Incomplete nutrition data.

13. Food Quantity

Food logging should support precise quantities.

The primary precise unit should be grams where appropriate.

Example: Jeera Rice - 200 g.

Support for millilitres should exist where relevant.

The system should not depend only on vague units such as 1 serving.

14. Food State

Food state must be explicitly represented because nutrition values can change substantially depending on preparation.

Possible states include:

- RAW.

- COOKED.
- PREPARED.
- DRY.
- UNKNOWN.

The UI should make this visible, preferably as a clear tag/badge.

Example: Chicken Breast - COOKED.

15. Food Search and Logging

Basic food logging flow: Dashboard -> Add Food -> Search -> Select food -> Enter quantity -> Select food state -> Calculate nutrition -> Review -> Confirm.

The user should not immediately persist a food entry simply by selecting a food.

16. Mandatory Food Review Before Logging

Every normal Phase 1 food log should go through a review step.

Before saving, show:

- Food name.
- Quantity.
- Unit.
- Food state.
- Meal/category.
- Provider nutrition.
- Calculated nutrition.
- Warnings/missing nutrients.

The user should then explicitly choose Confirm Log.

Only after confirmation should the historical FoodLog be created.

17. Nutrition Overrides Before Confirmation

The user should be able to manually adjust nutrition values before confirming the log.

Example: Provider calculates Protein = 10 g, but user changes it to 12 g.

The system should preserve:

- Provider Nutrition Snapshot - original values returned/calculated from the provider.
- User Override Nutrition - only values explicitly changed by the user.
- Final Nutrition Snapshot - final values used for the historical food log.

18. Historical Food Logs Must Be Stable

Once a food log is confirmed, its nutrition should represent what was recorded at that point in time.

Future provider changes should not modify historical nutrition records.

Example: provider says 100 g = 120 kcal today and later changes to 125 kcal; the old log should still retain 120 kcal unless explicitly edited by the user.

19. Editing Before Confirmation

Before confirming a food log, users should be able to change:

- Food.
- Quantity.
- Unit.
- Raw/cooked/prepared state.
- Meal/category.
- Nutrition values.
- Supported food attributes.

Changing quantity/state should normally trigger nutrition recalculation.

20. Custom Foods

Users should be able to create custom foods.

Example: My Homemade Protein Shake with manually entered nutrition information.

Custom foods should go through the same review/confirmation process when being logged.

21. Personal Serving Definitions

Basic personal serving definitions should be supported where practical.

Examples: 1 bowl = 250 g; 1 roti = 45 g.

This allows familiar quantities while preserving accurate gram-based nutrition underneath.

More advanced serving functionality can come later.

22. Meals

Users should be able to create reusable meals composed of multiple foods.

Example breakfast: oats, milk, banana, protein powder.

Meals should aggregate the nutrition of all included foods.

When logging a meal, the final nutrition summary should still be visible before confirmation.

23. Recent / Frequent / Favourite Foods

Users should be able to access:

- Recent foods.
- Frequently logged foods.
- Favourite foods.

24. Dashboard

The main dashboard should show:

- Current Tracking Day.
- Daily Reset Time.
- Selected dashboard nutrients.
- Food consumed today.
- Nutrition progress.
- Active nutrition targets.
- Medication/supplement summary.

Examples: Protein: 108 / 130 g; Calories: 1,750 / 2,300 kcal.

25. Medication and Supplement Tracking

Medication and supplements should use the same underlying HealthItem model.

Type:

- MEDICATION.
- SUPPLEMENT.

Fields may include:

- Name.
- Dosage / quantity entered by user.
- Unit.
- Instructions entered by user.
- Schedule.
- Start date.
- End date.
- Notes.
- Active/inactive status.

The application records what the user entered and should not prescribe treatment or dosage.

26. Medication/Supplement Scheduling

Phase 1 should support basic recurrence patterns such as:

- Daily.
- Alternate day.
- Weekly.
- Custom recurrence.

Example: Vitamin D - 1 tablet - Every alternate day.

27. Health Item Events

Possible statuses:

- SCHEDULED.
- TAKEN.
- MISSED.
- SKIPPED.
- RESCHEDULED.

This allows proper historical tracking rather than simply storing whether a medication exists.

28. Missed Medication Handling

A missed event should remain visible.

Examples: Missed yesterday; Pending event.

The application may surface missed events or allow configurable carry-forward behaviour.

However, it should never automatically instruct the user to:

- Double the dosage.
- Take additional medication.
- Make a medical decision.

It is a tracking system, not a prescription system.

29. History

Users should be able to inspect historical information for:

- Previous Tracking Day.
- Week.
- Month.
- Custom date range.

Exact food timestamps should still be retained even though daily aggregation uses Tracking Day.

Historical records should remain stable.

30. Phase 1 Summary

Phase 1 should provide:

- Authentication.
- Multiple users.
- User profile.
- Timezone.
- Configurable daily reset.
- Tracking Day abstraction.
- Nutrition requirement calculation.
- Reference requirements vs active targets.
- Target review.
- Manual target overrides.
- Target recalculation.
- Target history/versioning.
- Configurable dashboard nutrients.
- Full nutrition analysis foundation.
- External food provider integration.
- Indian food support.
- Gram/ml-based quantities.
- Raw/cooked/prepared food state.
- Mandatory food preview/review.
- Per-nutrient overrides.
- Provider/user/final nutrition snapshots.
- Explicit food confirmation.
- Custom foods.
- Meals.

- Recent/frequent/favourite foods.
- Personal serving definitions.
- Calories/macros tracking.
- Medication tracking.
- Supplement tracking.
- Basic schedules.
- Health-item event status.
- Missed-event tracking.
- Nutrition history.

31. Phase 2 - Advanced Nutrition and Health Tracking

Nutrition

- Comprehensive micronutrient tracking.
- Vitamins.
- Minerals.
- More detailed nutrient analysis.
- Nutrient drill-downs.

Recipes

- Ingredients.
- Quantity.
- Food state.
- Total nutrition.
- Serving count.
- Nutrition per serving.

Additional Phase 2 features

- Barcode scanning.
- Expanded Indian food support.
- Explicit water tracking.
- Weight history.
- Activity/exercise tracking.
- Advanced health-item scheduling and schedule history/versioning.
- Analytics such as weekly calorie averages, protein consistency, nutrient patterns, macro trends, and weight trends.
- Saved meals and richer personalized serving definitions.

32. Phase 3 - AI and Automation

Nutrition Label Recognition

- Upload or photograph a nutrition label.
- Extract calories, protein, carbohydrates, fat, micronutrients, and serving size.
- Always show extracted information for user review/editing before save.

Food Image Recognition

- Identify food.
- Estimate approximate quantity.
- Identify possible ingredients.
- Treat results as estimates and require user review/correction.

Natural-Language Food Logging

- Example: I ate 2 rotis, 150 g paneer curry and a bowl of curd.
- Convert into structured food entries and still require validation/review.

Voice Logging

- Use the same natural-language pipeline for voice input.

External AI / Agent Integration

- Query nutrition history.
- Add food logs.
- Retrieve nutrient totals.
- Create exports.
- AI systems should not bypass core validation/business rules.

33. MCP / External Interface

Possible permissions:

- nutrition.read
- nutrition.write
- history.read
- health.read
- health.write
- profile.read
- export.create

Permissions should be explicitly scoped.

34. Auditability

AI/external-agent operations should eventually support audit logs containing:

- Which agent performed the action.
- When.
- What data was changed.
- Previous value.
- New value.

35. Export

Future export formats may include:

- CSV.
- JSON.

This allows users to analyse or migrate their own data.

36. Explicitly Out of Scope / Deferred

Food Inventory Management

- No pantry inventory.
- No package count tracking.
- No grocery stock tracking.

Expiry Tracking

- No food-expiry management.

Automatic Medical Recommendations

- No diagnosis.
- No prescription.
- No treatment recommendations.
- No automatic medication adjustment.

Automatic Nutrition Recommendations

- Core app primarily tracks and displays information.
- External analytical/AI systems can be added later.

Native Mobile Application

- Phase 1 is browser/web based.

Offline Support

- Not required initially.

Push / Browser Notifications

- Deferred from the initial implementation.

37. Core Product Principle

Input -> Calculate/Retrieve -> Review -> User Confirmation -> Persist.

This is preferred over Input -> Automatically Persist, particularly for nutrition information.

The user remains the final authority over what is recorded in their nutrition history.

38. Medical / Reference Positioning

The product is intended for:

- Nutrition tracking.
- Medication tracking.
- Supplement tracking.
- Health-reference purposes.

It is not intended to provide:

- Diagnosis.
- Treatment.
- Prescription.
- Medical advice.

Nutrition requirements and calculated targets are reference values and may not represent an individual's medical needs.